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Recommendation Letter

This is to recommend Karine Kuyumzhiyan for participation in Summer School "*Structures in Lie Representation Theory*".

I know Karine since September 2003. She was among the best students in my algebra classes for the first and later for the second year students at Moscow State University.

Since September 2005, Karine is a student of Higher Algebra Department, and I am her scientific supervisor. Karine was proposed to study the problem of normality of maximal torus orbits' closures in irreducible modules of simple algebraic groups. It is well-known that a torus orbit closure is normal if and only if the corresponding set of weights satisfies a combinatorial condition called a saturation property. In order to find all irreducible modules such that the closure of any T-orbit is normal, one should list all weight systems with any subset being saturated. Some optimism concerning this problem was based on the following result due to J. Morand: in the adjoint module of the group $\mathrm{SL}(n)$ the closure of any T-orbit is normal. Karine find a complete solution of the problem for the group $\mathrm{SL}(n)$. This result is accepted for publication in Journal of Algebraic Combinatorics. Further, Karine studies the problem for other classical groups, and a complete classification is finished up to two cases not worked out yet.

Karine is an active participant of our seminars. Moreover, she attended winter and summer schools in Israel and France. These trips were useful for her education and scientific range of interests.

Since October 2008, Karine is a post-graduate student at Moscow State University and at Institute Fourier (Grenoble, France). Her scientific supervisors in the framework of this program are Professor Michail Zaidenberg and myself. During staying at Institute Fourier Karine combine her work on torus orbits with a new research devoted to automorphisms of affine varieties and locally nilpotent derivations.

In my opinion, Karine Kuyumzhiyan is a gifted and hard-working young mathematician. I think that she will fully benefit from attending the summer school, and I strongly recommend Karine for this event.